

Derbyshire Shared Care Pathology Guidelines

Myeloma Screening in Primary Care

A useful myeloma guide for GPs is available on the Myeloma UK website:

<https://www.myeloma.org.uk/wp-content/uploads/2021/09/Myeloma-UK-Myeloma-and-MGUS-A-Guide-for-GPs.pdf>

This guidance should be reviewed in conjunction with the Shared Care Pathology Guidelines for MGUS

When to screen for Myeloma and related disorders

- Not recommended to screen the normal population

The list below is not exhaustive, but a myeloma screen is strongly recommended when patients present with any of the following clinical symptoms or laboratory investigations.

- Clinical Symptoms:
 - Unexplained drenching night sweats
 - Persistent and unexplained bone pain, particularly in the back or the ribs for more than 4 weeks
 - Recurrent Infections
 - Unexplained peripheral neuropathy or bilateral carpal tunnel syndrome
 - Unexplained significant weight loss
- Any of the following pathology findings with no known explanation:
 - ESR > 30 mm/h
 - Adjusted Calcium > 2.65 mmol/L (PTH low, and no known Primary Cancer)
 - New renal impairment, unexplained AKI and/or nephrotic range proteinuria
 - Unexplained anaemia
 - Abnormal X ray findings such as lytic lesions, osteoporotic vertebral fractures
- Please note that depending on the clinical details provided, the lab will aim to perform protein electrophoresis on samples that we detect with:
 - high or low globulins without a known cause
 - hypercalcaemia and a suppressed PTH without a known cause
- At the moment the laboratory does not automatically undertake a myeloma screen on new, unexplained cases of anaemia, raised ESR or renal impairment.

What is a Myeloma Screen?

The basic Myeloma Screen consists of the following tests

FBC

UE

Adjusted Calcium

Serum Protein electrophoresis (SPE)

Both a serum sample (either red or gold top) **and** an EDTA sample (Purple top) are required.

Serum Protein Electrophoresis (SPE) can detect an unusual protein band in the serum (a monoclonal band). These bands are also known as paraproteins
Serum free light chains (SFLC), Immunoglobulins and immunofixation will be added by the lab as appropriate to the SPE findings/clinical details.

Serum Protein Immunofixation

This will determine whether there is truly a monoclonal band (paraprotein) present and identify the type of band present if there is an abnormality noted on serum protein electrophoresis..

When a paraprotein is detected, the isotype will be identified in the report (eg IgG kappa, IgA lambda etc).

These are usually solitary protein bands, but 2 or more bands are sometimes present, sometimes of the same heavy and light chain type, or they can be of different types.

Sometimes the only detectable abnormality is an elevated light chain without the presence of a paraprotein.

Monoclonal Protein Quantitation

It should be noted that the level of the paraprotein is a distinct measurement from the total immunoglobulins level. For referral purposes, the level of the paraprotein should be used rather than the total immunoglobulin level.

Measurement of Total Major Immunoglobulins

Total IgA, IgG and IgM are determined

- For individuals with reduced Igs with no abnormal paraprotein or SFLC ratio and with no clinical suspicion of myeloma/lymphoma and no lymphocytosis, consider taking advice from an immunologist
- Note that IgG levels are often lower in individuals >60 yr and an isolated reduced IgM is often not significant

Measurement of Free Light Chains

The quantities of free kappa and free lambda light chains are determined in the serum and the light chain ratio is then determined. If there is free light chain secreting monoclonal gammopathy, the light chain ratio can become abnormal.

Adult Reference Ranges are:

	Age <70 yrs	Age ≥70 years
Free kappa	6.3- 39.0 mg/L	7.0-55.8 mg/L
Free lambda	5.9 - 36.7 mg/L	6.0-48.0 mg/L
Kappa/Lambda ratio	0.44 - 2.16	0.46 - 2.59

- Many situations can give rise to a combination of abnormally high free kappa and free lambda concentrations, with a normal or only mildly abnormal ratio. This is typically not of any significance and **does not require monitoring**
- Interpretive comments are added to FLC results to aid decisions about the need to repeat/monitor
- Small increases in the FLC ratio (up to 3.38) can be seen in renal impairment or polyclonal increases in immunoglobulins, and are not usually clinically important

Myeloma screen result outcomes

The results from the myeloma screen and any tests reflexed are interpreted the laboratory and guidance issued in the report as to whether further action is required. There are essentially two categories of outcome

1. No apparent paraprotein detected

There are multiple scenarios that fall into this category, the most common of which are listed in the table below

Protein electrophoresis result	Free light chains	Further action required?
No apparent paraprotein	Within reference ranges	No
No apparent paraprotein	Individual light chains raised but ratio within reference range	No
No apparent paraprotein in patient with CKD	Free light chain ratio raised but within range expected for renal impairment	No
Polyclonal increase in globulins	Not added or within reference range	No
Polyclonal increase in globulins	Individual light chains raised but ratio within reference range	No
Polyclonal increase in globulins	Free light chain ratio raised but within range expected for polyclonal increase	No
Hypogammaglobulinaemia	Within reference ranges	Consider advice from Immunologist
Hypogammaglobulinaemia	Individual light chains raised but ratio within reference range	Consider advice from Immunologist

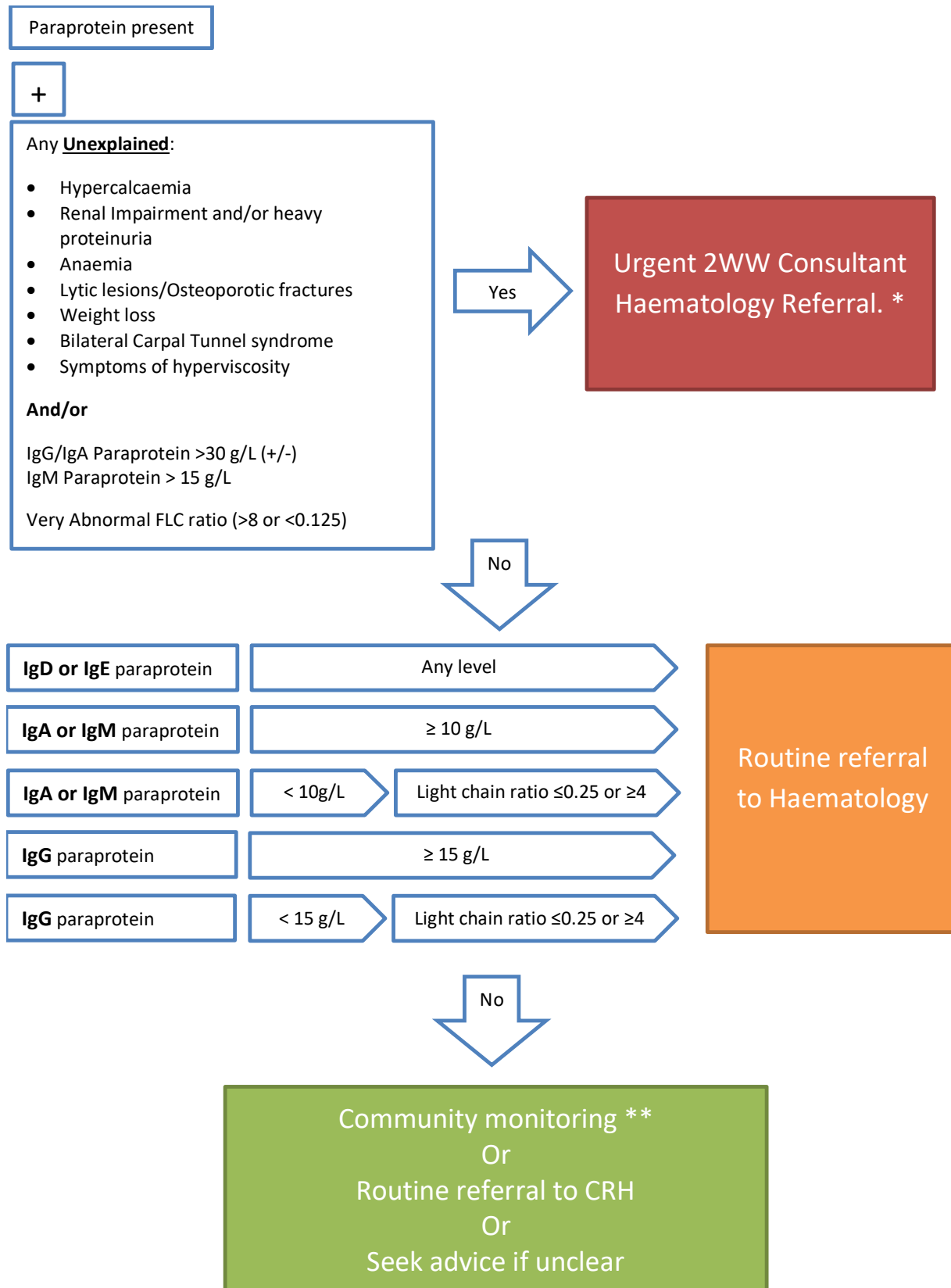
Scenarios that fall outside of these will be interpreted on a case by case basis and appropriate guidance added by the laboratory.

2. Paraprotein detected

The laboratory will interpret and add interpretative guidance based upon the flow chart below 'Referral criteria for newly identified paraproteins' including advice on referral.

It is important to view this interpretation in clinical context and the ultimate responsibility of making a referral to the haematologist lies with the primary care physician.

Referral criteria for newly identified paraproteins. For more information see specific MGUS shared Care pathology Guideline



*If new/significant renal impairment consider renal referral in addition.

**For further information see shared care guideline for MGUS

MGUS

MGUS is the commonest cause for the presence of a paraprotein in a patient's serum and is a relatively common disorder in the elderly population, with a low rate of progression to malignant disease (~1% per year). Therefore, the majority of patients with MGUS never need treatment. The size and type of paraprotein, in addition to other factors can help to determine which patients require Consultant Haematology referral and which patients can be managed in the community.

For further information on the management of patients with MGUS and when to refer to a Consultant haematologist please refer to the **shared care guideline for MGUS**.

Useful Contacts

UHDB Consultant Haematologist	01332 787973 (secretary)
UHDB Clinical Nurse Specialist	07825 342918
CRH Consultant Haematologist	crhft.cabadviceandguidance@nhs.net
QHB Consultant Haematologist	01283 511511 ext. 4025 (Secretary)
UHDB Duty Biochemist	01332 789383
CRH Duty Biochemist	01246 512212

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